Rain Prediction Methods as stated in the Kṛṣiparāśara

Introduction: Rain prediction was crucial not only for agriculture but also for daily livelihood, in both modern and ancient times. There were many efforts in ancient civilisations to predict this phenomenon. Two methods are employed here: the observational method and the calculation method. Observational methods are also of two kinds: natural observation and observation of the results of certain rituals. Natural observation encompasses wind direction, wind flow patterns, rain patterns, astronomical and sky type observations, and thunderstorm activity. Although the Gaṇita Jyotiṣa method, which falls under the calculation methods, is widely used, the observation method is also useful at times, particularly for understanding sudden rainfall. Parāśara himself gives a beautiful verse on why the knowledge of the rain prediction is so important for agriculture:

वृष्टिमूला कृषिः सर्वा वृष्टिमूलञ्च जीवनम्।
तस्मादादौ प्रयत्नेन वृष्टिज्ञानं समाचरेत्।।

It has been stated that Farming depends wholly on rainfall. Life itself. Therefore, depends on rainfall. Hence, to begin with, one should strive hard to acquire the knowledge of rainfall.

Kṛṣiparāśara Methods: There are four specific methods for predicting rain in the kṛṣiparāśara, based on the ruler planet, numerical calculations for the year, the outcome of the ritual, the movement of celestial bodies, and natural entities. In this text, there are only seven planets (*graha*). So, the seven-year cycle comes into existence here.

The Method of finding out the ruler (*graha*) of the year:

शाकं त्रिगुणितं कृत्वा द्वियुतं मुनिना हरेत्।
भागशिष्टो नृपो ज्ञेयो नृपान्मन्त्री[1] चतुर्थकः।। 12

Multiply the number denoting the śaka year[2] by three and add two. Divide the result by seven (The number of the sages). The remainder indicates the ruling planet for that

[1] There are multiple interpretations of this नृपान्मन्त्री (नृपात् मन्त्री): In Nalini Sadhale's ed., it is fifth from the ruler planet, ed. of Dwarka Prasad Shastri's is fourth from the ruler.

[2] The verse is mentioning the śaka year but as there are two Shaka era systems in scholarly use, one is called Old Shaka Era, which is uncertain, probably sometime in the 1st millennium BCE because ancient Buddhist and Jaina inscriptions and texts use it as well as in the southern part of India, but this is a subject of dispute among scholars. The other is called śaka era of 78 CE, or simply śaka era, a system that is common in epigraphic evidence from southern India. A parallel northern India system is the Vikrama era, which is used by the Vikram calendar linked to Vikramāditya.

Saka year. The earth, Fourth from the ruling planet, means the minister planet of that year.

So, here is the step-wise format:

Every year, a particular planet is the ruler, and another planet is the minister.

- Multiply the Saka year number by 3 and add 2.
- Divide the result by seven (the number of sages).
- The remainder indicates the ruling planet of that Saka year.
- The planet, which is fourth from the ruling planet, indicates the minister planet of that year.

To explain this formula, the following illustration may be helpful. Let the śaka year be 1920.

$1920 \times 3 = 5760$
$5760 + 2 = 5762$
$5762 \div 7 = 823 +$ remainder 1

The main "planets" (old concept) are listed in the following order: 1. Sun, 2. Moon, 3. Mars, 4. Mercury, 5. Jupiter, 6. Venus, 7. Saturn.

Therefore, the ruling planet of the 1920 Saka year is the Sun. The "minister planet" of the year is Venus, as it is the fifth planet from the Sun.

The result of the particular *graha* as the ruler of the year:

The general results

चित्तलार्के नृपे वृष्टिर्वृष्टिरुग्रा निशापतौ ।
वृष्टिर्मन्दा सदा भौमे चन्द्रजे वृष्टिरुत्तमा ॥13 ॥
गुरौ च शोभना वृष्टिर्भार्गवे वृष्टिरुत्तमा ।
पृथिवी धूलिसम्पूर्णा वृष्टिहीना शनौ भवेत् ॥14 ॥

As per the calculation written above:

- The sun, as the year's ruler, indicates average rainfall,
- the Moon heavy rains, Mars scanty rains, and Mercury good rains.

- When Jupiter happens to be the king of the year, the rainfall is satisfactory.
- Venus indicates excellent rainfall,
- while Saturn, as a king, leaves the earth dry and dusty.

Specific results based on the *graha*s as the ruler

The result when Sun is the ruler:

चक्षुरोगो ज्वरादिष्टं सर्वोपद्रव एव च।
मन्दावृष्टिः सदावातो यत्राब्दे भास्करो नृपः॥ 15॥

When ruled by the Sun in that year, it causes various calamities, including eye diseases, fever threats, scant rainfall, and continuous windstorms.

The result when moon is the ruler:

यस्मिनसंवत्सरे चैव चन्द्रो राजा भवेद्ध्रुवम्।
कुर्यात्सस्यान्वितां पृथ्वीं नैरुज्यं चापि मानवे॥ 16॥

In which year the moon is the ruler is for sure, to enrich the earth with good harvest and bestow health on mankind.

The result when Mars is the ruler:

शस्यहानिर्भवेत्तत्र नित्यं रोगश्च मानवे।
यस्मिन्नब्दे कुजो राजा शस्यशून्या च मेदिनी॥17॥

In the year ruled by Mars, the crops and diseases spread among people. The earth becomes bereft of crops.

The result when the murcury is the ruler:

नैरुज्यं सुप्रचारश्च सुभिक्षं क्षितिमण्डले।
यत्राब्दे चन्द्रजो राजा सर्वशस्या च मेदिनी॥18॥

When Mercury is the ruler, the earth is free of diseases. Transportation is easy, and there is plenty of harvest. The world is blessed with a wide variety of crops.

The result when the Jupiteris the ruler:

धर्मस्थितिर्मनःस्थैर्य वृष्टिकारणमुत्तमम्।
यस्मिन्नब्बे गुरू राजा सर्वा वसुमती मही॥ 19॥

If Jupiter rules the year, Dharma prevails on the earth. People enjoy peace of mind. There is good rainfall. The whole world enjoys prosperity.

The result when the Venus is the ruler:

नृपाणां वर्धनं नित्यं धनधान्यादिकं फलम्।
राजा दैत्यगुरुः कुर्यात् सर्वशस्यं रसातलम् ॥२० ॥

Venus (the preceptor of demons), as a ruler of the year, causes the kings to prosper without fail: prosperity and plenty result. The earth is blessed with a variety of food grains.

The result when the Saturn is the ruler:

संग्रामो वातवृष्टिश्च रोगोपद्रव एव च।
मन्दावृष्टिः सदा वातो नृपे संवत्सरे शनौ ॥२१ ॥

The year in which Saturn rules is associated with war, stormy rains, and outbreaks of disease. Rains are scanty, and winds are continuous.

Ending Remarks

यथा वृष्टिफलं प्रोक्तं वत्सरग्रहभूपतौ।
तद्वद्दृष्टिफलं ज्ञेयं विज्ञैर्वत्सरमन्त्रिणि ॥२२ ॥

Experts should speculate rainfall concerning the 'minister planet' of the respective year in the same manner as stated above regarding the king of the year[3].

<u>The method of ascertaining the type of cloud of the Year:</u>

शकाब्दं वह्निसंयुक्तं वेदभागसमाहृतम्।
शेषं मेघं विजानीयादावर्तादि यथाक्रमम्।। 23

By adding the number of types of fire (3) to the number denoting the Śaka Year and dividing that sum by the number of the Vedas (4), with the remainder (in this calculation), the type of cloud should be identified, as per the sequence starting from the Āvarta (listed in the next).

आवर्तश्चैव संवर्तः पुष्करो द्रोण एव च।
चत्वारो जलदाः प्रोक्ता आवर्तादियथाक्रमम्।। 24

[3] This is kind of a seven-year cycle, as the number of the *graha*s is 7 here.

Āvarta (Cumulonimbus[4]), Saṁvarta (Altostotrus[5]), Puṣkara (Vortex[6]) and Droṇa (Stratocumulous[7]) are the four types of cloud (Jaladā), which have been said (by sages) in the sequence from Āvarta (Talked about before).

एकदेशेन चावर्तः संवर्तः सर्वतो जलम् ।
पुष्करे दुष्करं वारि द्रोणे बहुजला मही ।। 25

(Rains) Some parts not everywhere (are called) Āvarta, Saṁvarta is the (condition when) everywhere water (may be seen through water). In the case of Puṣkara (cloud), water (rain) is scanty, and Droṇa (indicates) the earth (the place where the cloud could be seen) gets full of water.

If we go step by step:

Āvarta, Saṁvarta, Puṣkara and Droṇa are the four types of cloud (Jaladā). Āvarta is the first in order.

Method:

- Add the number of types of fire (which is three) to the number denoting the Saka year (i.e., 1920).
- Divide the sum by the number of the Vedas (which is four).
- The remainder of the division (in this calculation, 1+9+2+0=12/4=3) indicates the type of the cloud, viz., Aavarta, etc., according to their order.

Characteristics of clouds:

- Aavarta rains in one part, while Samvarta rains everywhere.
- Water is very little when Pushkara is the cloud of the year, but Drona makes the earth full of water.

<u>The Method of determining the amount of rainfall:</u>

शतयोजनविस्तीर्णं त्रिंशद्योजनमुच्छ्रितम् ।

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[4] Cumulonimbus gives a thunderstorm in a short span of time and in a limited space. special feature of its buildup in the sky with a base at about 2500-3000 ft (about 750-900 m) above ground and with a vertical ascent of 25,000-30,000 ft (about 7-9 km) above its base. (Balkundi)

[5] An altostratus, which is widely spread in the sky, at a height from 2.5 km to 6.0 km above sea level. The thickness of the sheet-cloud can be considerable, rendering the Sun invisible during the period of its spread in the sky. (Balkundi)

[6] A vortex, shows that it is a cloud of short duration as its buildup is a temporary phenomenon or a disturbance in the normal atmosphere. (Balkundi)

[7] stratocumulus which is a sheet-cloud at a height of approximately 2 km above ground. This type also gives steady, continuous rain. (Balkundi)

आढकस्य भवेन्मानं मुनिभिः परिकीर्तितम्।।26

(Water quantity contained in the place with) 100 Yojana[8] large and 30 Yojana of depth is the measurement of Āḍhaka, as stated by the sages.

युग्माजगोमत्स्यगते शशाङ्के रविर्यदा कर्कटं प्रयाति।
जलं शताढं हरिकारमुकेर्द्धं वदन्ति कन्यामृगयोरशीतिम्।।27

When the Moon (Śaśāṅka) enters Gemini (Yugma), Aris (aja) or Taurus (Go) or the Pisces (Matsya) and the sun (Raviḥ) enters the Cancer (Karkaṭa), the rainfall (Jala) would be (of the) 100 Āḍhaka (quantity). Suppose the Sun passes through Leo (Hari) and Sagittarius (Kārmuka), it is half and when Virgo (Kanyā) or Capricornus (Mr̥ga), it is 80 (Āḍhakas).

कुलीरकुम्भालितुलाभिधाने जलाढकं षण्णवतिं वदन्ति।
अनेन मानेन तु वत्सरस्य निरूप्य नीरं कृषिकर्म कार्यम्।।28

When the (Sun) enters Cancer (Kulīra), Aquarius (Kumbha), Scorpio (āli), Libra (Tulā), The Water (from rain) is 96 āḍhaka, (as) said by many. Calculating water measurements throughout the year enables agricultural tasks to be accomplished.

समुद्रे दशभागांश्च षड्भागानपि पर्वते।
पृथिव्यां चतुरो भागान् सदा वर्षति वासवः।।29

The Vāsava or Lord īndra (allocate) rains- ten parts (of the water) in the ocean, six parts to the mountain and four to the earth (other than the earlier two parts).

If we go step by step

- The Vāsava or Indra (the rain-god) allocates ten parts (Daśabhāgān) of rainwater to the ocean, six parts (Ṣaḍbhāgān) to the mountain (Parvate) and four parts (Caturo Bhāgān) to the earth (other than the earlier two parts).

Method:

- Sages have fixed Āḍhaka (Four Korashas = One Yojana[9]) as the measure of water, which is the quantity of water contained in an expanse of a hundred yojanas (Śatayojana-Vistīrṇaṁ) large and the depth of thirty yojanas (Triṁśadyojanamucchritaṁ).
- When the Sun (Ravi) enters the sign of Cancer (Karkaṭa) while the Moon (Śaśāṅka) is in Gemini (Yugma), Aries (aja), Taurus (Go), or Pisces (Matsya), the

[8] Though there are variations in the actual measurement of the Yojana ranges ≳ 7.31- ≲ 15 km.
[9] Monnier Williams

rainfall would be a hundred Āḍhas (Śatāḍhaṁ), and if the Sun passes through Leo (Hari) and Sagittarius (Kārmuka), it is fifty Āḍhakas (Ardhaṁ).

- When the Sun is in Virgo (Kanyā) or Capricornus (Mr̥ga), rainfall is stated to be eighty (Aśīti) Adhakas.
- When the Sun enters Cancer (Kulīra), Aquarius (Kumbha), Scorpion (āli), Libra (Tulā), the rainfall is 96 Āḍhaka (ṣaṇṇavati).

Ending Remarks:

- Calculating with this measurement of water (Nīrupya Nīraṁ) throughout the year (Vatsarasya), the task for agriculture should be done (Kr̥ṣikarma Kāryaṁ).

<u>Prediction of the rainfall in the month of Pauṣa (Based on the wind direction)</u>

सार्द्धं दिनद्वयं मानं कृत्वा पौषादिना बुधः।
गणयेन्मासिकीं वृष्टिमवृष्टिं वानिलक्रमात्।।30

Sārdhaṁ (One and a half) Dinadvayam (Two Days) Mānaṁ (Measurement) Kr̥tvā (by doing) Pauṣādinā (in the Pauṣa etc.) Budhaḥ (Wise person). Gaṇayet (Shall be counted) Māsikīṁ (Monthly) Vr̥ṣṭiṁ (Rain) Vānilakramāt (By the sequence/ Nature of the air flow).

सौम्यवारुणयोर्वृष्टिरवृष्टिः पूर्वयाम्ययोः।
निर्वाते वृष्टिहानिः स्यात् संकुले संकुलं जलम्।।31

Saumya-vāuṇayoḥ (in the North and west) Vr̥ṣṭi (Rain) Avr̥ṣṭi (absence of the rain) Pūrva-Yāmyaoḥ (in the East and the South). Nirvāte (In the absence of air flow) Vr̥ṣṭihāniḥ (loss of rain) Syāt (would be) Saṁkule (in irregularity- or wind) saṁkulaṁ (irregularity) jalaṁ (of the water).

एकेकं पञ्चदण्डेन मासस्य दिवसो मतः।
पूर्वार्द्धं वासरी वृष्टिरुत्तरार्द्धे च नैशिकी।।32

Ekaikaṁ (Every one part) Pañcadaṇḍena (of the five Daṇḍas) Māsasya (of a month) Divasaḥ (one day) mataḥ (considered). Pūrvārdhe (in the first half) Vāsarī-Vr̥ṣṭiḥ (Rain in the daytime) Uttarārdhe (in the second half) ca (and) Naiśikī (in the night).

दत्त्वा दण्डे पताकां तु वातस्यानुक्रमेण च।
विज्ञेया मासिकी वृष्टिः कृत्वा यत्नमहर्निशम् ॥३३॥

Datvā (By putting) Daṇḍe (in the stick) Patākāṁ (the flag) tu (indeed) Vātasyānukrameṇa (according to the air-flow direction) Vijñeyā (Should be known)

Māsikīṁ (Monthly) Vr̥ṣṭiḥ (Rain) Kr̥tvā (by doing) Yatnamaharniśaṁ (efforts day and night)

धूलीभिरेव धवलीकृतमन्तरिक्षंविद्युच्छटाच्छुरितवारुणदिग्विभागम्।
पौषे यदा भवति मासि सिते च पक्षे तोयेन तत्र सकला प्लवते धरित्री ॥३४॥

Dhūlibhireva (by the dust only) Dhavalīkr̥tamantarikṣaṁ (whitened sky) vidyucchaṭācchurita (lightened with the ray of the thunder/lightning) Vāruṇadigvibhāgaṁ (the western side), Pauṣe (in the month of Pauṣa) yadā (when) bhavati (happened) māsi (in the month) site pakṣe (in the bright half of the month), Toyena (with the water) Tatra (there) sakalā (full) Plavate (gets flooded) dharitrī (the earth).

पौषे मासि यदा वृष्टिः कुज्झटिर्वा यदा भवेत्।
तदादौ सप्तमे मासि वारिपूर्णा भवेन्मही ॥३५॥

Pauṣe Māsi (in the month of Pauṣa) yadā (when) Vr̥ṣṭi (rain) Kujjhaṭirvā (or the fog) yadā (when) Bhavet (will happen), Tadādau (in that from this) saptame Māsi (seventh month) Vāripūrṇā (full of water) Bhavenmahī (the earth will be).

यदा पौषे सिते पक्षे नभो मेघावृतं भवेत्।
तोयावृता धरित्री च भवेत् संवत्सरे तदा ॥३६॥

Yadā Pauṣe (when in the month of Pauṣa) Site Pakṣe (at the bright half) Nabhaḥ (the sky) Meghāvr̥taṁ Bhavet (gets covered by the cloud). Toyāvr̥tā (covered with the water) Dharitrī (the earth) ca (and) Bhavet (will be) saṁvatsare (in that year) tadā (that time).

मीनवृश्चिकयोर्मध्ये यदि वर्षति वासवः।
तदादौ सप्तमे मासि तत्तिथौ प्लवते मही ॥ ३७ ॥

Mīna-Vr̥ścikayormadhye (in between the Mīna and Vr̥ścika) yadi (if) Varṣati (rains) vāsava (Lord Īndra) Tadādau (at first) saptame māsi (in the seventh month) tattithau (in that tithi) Plavate (gets flooded) Mahī (the Earth).

So, if we see step-by-step:

- An expert (Budhaḥ) should measure (Mānaṁ) rainfall or lack of it every month (Māsikīṁ Vr̥ṣṭiṁ) starting with *Pauṣa* by studying the direction of the winds (Vātānilakramāt), etc. Taking (Kr̥tvā) two and a half days as a unit (Sārdhaṁ dinadvayaṁ Mānaṁ).
- Wind direction from north and west indicates rainfall, and that from the east and south indicates the absence of Rainfall.

- If there is no movement of wind in the atmosphere indicates the loss of Rainfall, and irregular movement of wind indicates irregular Rainfall.
- Five Dundas is the unit of a day for the month. If the wind blows in the first half of the month, it rains during the day; in the second half, it rains during the night.
- To determine monthly rainfall, one should diligently track the wind day and night, keeping a rod with a flag attached to mark its direction.
- When the atmosphere is full of dust, and the western direction is fragmented by flashes of lightning in the bright half of Pausha, the entire earth is sure to be drenched with water.
- If in *Pauṣa* of a certain year there is rainfall and fog, the earth is likely to be inundated (Watery area) in the seventh month since then (i.e., July/August).
- If the bright half of *Pauṣa* sky is overcast with clouds, the earth becomes inundated in that year too.
- If Indra (Vāsava), the rain-god, causes rainfall in the middle of Pisces (Mina) (February-March) and Scorpio (Vr̥ṣcika) (October-November), the earth is drenched on the same lunar day (tithi) in the seventh month since then.

<u>Prediction of the rainfall in the month of Māgha (Based on the wind flow and thunderstorm)</u>

माघे बहुलसप्तम्यां तथैव फाल्गुनस्य च।
चैत्रे शुक्लतृतीयायां वैशाखे प्रथमेऽहनि ॥ ३९ ।।

Māghe (in the month of Māgha) Bahulasaptamyāṁ (in the Seventh Tithi of the dark half) Tathaiva (also) Phālgunasya (of the month of Phālguna) ca (and). Caitre (in the month of Caitra) Śuklatr̥tīyāyāṁ (in the third tithi of the bright half part) Vaiśākhe (in the Vaiśākha) Ahani (at the daytime).

एतासु चण्डवातो वा तडिद्वृष्टिरथापि वा।
तदा स्याच्छोभना प्रावृड् भवेत् शस्यवती मही ॥४० ॥

Etāsu (in these) Caṇḍavāta (Stormy Wind) vā (or) Taḍit-Vr̥ṣṭiḥ (Thunder shower) Athāpi (and also) Vā (or). Tadā (then) Sācchobhanā Prāvr̥ḍ (would happen good rain) Bhavet (should be) sasyavatī (with good crop) Mahī (Earth).

सप्तम्यां स्वातियोगे यदि पतति जलं माधपक्षेऽन्धकारे
वायुर्वा चण्डवेगः सजलजलधरो गर्जितो वासवो वा।
विद्युन्मालाकुलं वा यदि भवति नभो नष्टचन्द्रार्कतारम्
तावद्वर्षन्ति मेघा धरणितलगता यावदाकार्तिकान्तम् ॥४१ ॥

Saptamyāṁ (in the seventh tithi) Svātiyoge (in the Svāti constellation) yadi (if) Bhavati (happens) nabhaḥ (the sky) naṣṭacandrārktāraṁ (invisibility of the sun, moon and other stars), Tāvadvarśanti (till the time it rains) meghāḥ (the Cloud) Dharaṇitalagatā (came to the surface of the earth) Yavākārtikāntaṁ (up to the time of the end of the Kārtika Month).

माघे मासि निरन्तरं यदि भवेत् प्रालेयतोयागमो
वाता वान्ति च फाल्गुने जलधरैश्चित्रे च छन्नं नभः ।
वैशाखे करकाः पतन्ति सततं ज्यैष्ठ प्रचण्डातपाः
तावद्धर्षति वासवो रविरसौ यावत्तुलायां व्रजेत् । ।४२ ॥

Māghe Māsi (in the month of Māgha) Nirantaraṁ (Continuously) yadi (if) Bhavet (should happen) Prālayatoyāgamaḥ (Coming of the rain with the destruction/Thunder)

Vātā Vanti (The wind blows) ca Phālgune (in the Phālguna month) Jaladharaiḥ (with the Cloud), Citre (in the star month Chaitra) Channaṁ nabhaḥ (Overcusted Sky).

Vaiśāhe (in the month of Vaiśākha) Karakāḥ (the hail) Patanti (Fall). Satataṁ (indeed) Jaiṣṭhye (in the month of Jaiṣṭhya) Pracaṇḍātapaḥ (Scorching Heat). Tavat Varṣanti (Till then Rains) Vāsava (Lord Indra) Ravirasau (this Sun) Yāvat Tulāyāṁ Vrajet (Upto when gets transit into the Tulā)

So, if we see step-by-step:

- If on the seventh day of the bright half of Māgha it rains or clouds appear in the sky, the year brings with its blessings in the form of crops of all kinds.
- The rainy season is satisfactory, and the earth yields sufficient crop if stormy winds or thunder showers are noticed on the seventh day of the dark half of Māgha or Phalguna or on the third day of the bright half of Chaitra or on the first day of Vaisakha.
- Clouds hang low on earth bringing rains right up to the end of Kartika (November) in a year in which on the seventh day of the dark fortnight of Magha, (The rain happens in the yoga of swati constellation), if winds blow with high velocity; if the sky is laden with rain clouds; if there is rain with thunder; or if the sky is agitated with continuous flashes of lighting resulting in the invisibility of the Sun, the Moon and the Stars.
- If in the month of Māgha, it rains continuously with presence of sleet or if in Phalguna (March) winds blow with the clouds and if in Chaitra (April) the sky remains overcast and if in Vaisakha (may) there is a continuous hail shower and if in Jyeshta (June) there is scorching heat, the rain-god Indra causes rainfall right until the Sun enters the sign of Libra.

Prediction of the rainfall in the month of *phālguna* (Based on the Astronomical position)

पश्चम्यादिषु पञ्चसु कुम्भेऽर्के यदि भवति रोहिणीयोगः।
अधमतमाधममध्यममहदतिमहाम्भसां निपातः ॥४३॥

If on any one of the five days (Pañcasu) beginning with (ādiṣu) the fifth (Pañcamī), there is a coincidence (yogaḥ) of the Sun being (Arke) in the sign of Aquarius (Kumbhe) and also in the Rohini constellation, the rainfall (Ambhasāṁ nipātaḥ) in the year is very scanty (Adhamatama), scanty (Adhama), average (Madhyam), profuse (Mahad), and excessive (Atimahā), respectively.

Prediction of the rainfall in the whole year (Based on the rain observation at *caitra*)

प्रतिपदि मधुमासे भानुवारः सितायां
यदि भवति तदा स्याश्चित्तला वृष्टिरब्दे।
अविरलपृथुधारासान्द्रवृष्टिप्रवाहै-
र्धरणितलमशेषं प्लाव्यते सोमवारे ॥४४॥

If the first day of the bright fortnight of *Caitra* (Madhumasa) happens to be Sunday, rainfall in the year is moderate, and if it happens to be Monday, the entire surface of the earth is drenched with water flowing due to continuous, forceful, and heavy rains.

अवनितनयवारे वारिवृष्टिर्न सम्यग्
बुधगुरुभृगुजानां शस्यसम्पत्प्रमोदः।
जलनिधिरपि शोषं याति वारे च शौरे-
र्भवति खलु धरित्री धूलिजालैरदृश्या ॥४५॥

If the first day of the bright fortnight of *Caitra* happens to be a Tuesday, rainfall is not satisfactory in that year. If it is a Wednesday, Thursday, or Friday, good crops and happiness are indicated by good rain. However, if it happens to be a Saturday, even the ocean will dry up, and the earth will indeed become invisible due to the mantle of dust in the atmosphere.

Prediction of the rainfall in the month of *Caitra* (Based on the rain pattern observation Astronomical position)

चैत्राद्यभागे चित्रायां भवेच्च चित्तला क्षितिः।
शेषे नीचैर्न वात्यर्थं क्ष्मामध्ये बहुवर्षिणी ॥४६॥

If the first half of the month (Pakṣa) of *Caitra* in the citrā nakṣatra stays, then the rain would be moderate; in the last part (or the second part), it doesn't rain very much, and in the middle, it rains a lot.

मूलस्यादौ यमस्यान्ते चैत्रे वायुरहर्निशम्।

आर्द्रादीनि च ऋक्षाणि वृष्टिहेतोर्विशोधयेत् ॥४७॥

If in the month of *Chaitra*, at the start of the Mūla and end of the Yama (Bharaṇī)- the wind blows day and night, then the stars like Ārdrā etc. should be considered for searching for the cause of the rain (Rain would definitely happen).

<u>Prediction of the rainfall in the month of *vaiśākha* (Based on the ritualistic result and astronomical findings)</u>

प्रवाहयुतनद्यां तु दण्डं न्यस्य जले निशि।
वैशाखशुक्लप्रतिपत्तिथौ वृष्टिं निरूपयेत् ॥४८॥

Rains should be observed by putting at night a rod in the water of a flowing river on the first day of the bright half of the month.

औं सिद्धिरिति मन्त्रेण मन्त्रयित्वा शतद्वयम्।
अङ्कयित्वा तु तद्दण्डमङ्कतुल्ये जले क्षिपेत् ॥४६॥

With the spell 'Auṁ siddhi' chanted two hundred times, the stick should be marked, and the stick should be placed (lit. Thrown) at the same water level (matches the marking).[10]

प्रातरुत्थाय सहसा तदङ्कं तु निरूपयेत्।
समं चैवाधिकं न्यूनं भविष्यजलकाङ्क्षया ॥५०॥

Upon rising in the morning, one should immediately observe the marking and determine if the water level is the same, raised, or reduced, thereby knowing the amount of rainfall for the year.

<u>Prediction based on the result from the Ritural</u>

[10] [What is intended here seems that the rod should be firmly implanted in the river bed and a mark should be made on it at the water level.]

गतवत्सरवद्वारि वन्या चैव समे भवेत्।
हीने हीनं भवेद्वारि भवेद् बन्या च तादृशी ॥५१॥

If the level is the same, water and rainfall will be the same as in the previous year. If rainfall levels are reduced, water will be less than in the previous year.

अङ्काधिक्ये च द्विगुणा वृष्टिर्वन्या च जायते।
इति पराशरेणोक्तं भविष्यद्वृष्टिलक्षणम् ॥५२॥

If the level exceeds the mark, rainfall and water will double in volume. These are the criteria for forecasting rainfall as stated by Parāśara.

Method based on astronomical observation (*vaiśākha*)

सूर्योदये विषुवतो जगतां विपत्तिर्मध्यं गते दिनकरे बहुशस्यहानिः।
अस्तं गते दिनकरे तु तदर्द्धशस्यं ऐश्वर्यभोगमतुलं खलु चार्द्धरात्रे ॥५३॥

If the Sun is at the equinoctial point at sunrise, calamity will befall on the world; if it does so at midday, a huge loss of crops is indicated, and if it is at sunset, the harvest will be only half of that of the previous year; if it is at midnight, unmatched prosperity and happiness are indicated.[11]

Another Ritualistic method (*vaiśākha*)

रेखातयं समुल्लिख्य ताभिस्ताश्च विवर्द्धयेत्।
त्रिशृङ्गं सर्वकोणेषु पर्वतं तत्र दापयेत् ॥५४॥

Having drawn three lines, they should be multiplied by the same number. A three-peaked 'mountain' should then be drawn at each corner.[12]

ईशानादिदक्षिणाङ्कान् संलिखेदनलादितः।
येन येनाजसंक्रान्तिस्तेन प्रावृट्फलं भवेत् ॥५५।।

Starting with the northeastern (triangle) numbers denoting the nakshatras should be serially written in a clockwise direction (from īśāṇa to dakṣiṇa), the first being that which denotes Anala, i.e., Krittika.[13]

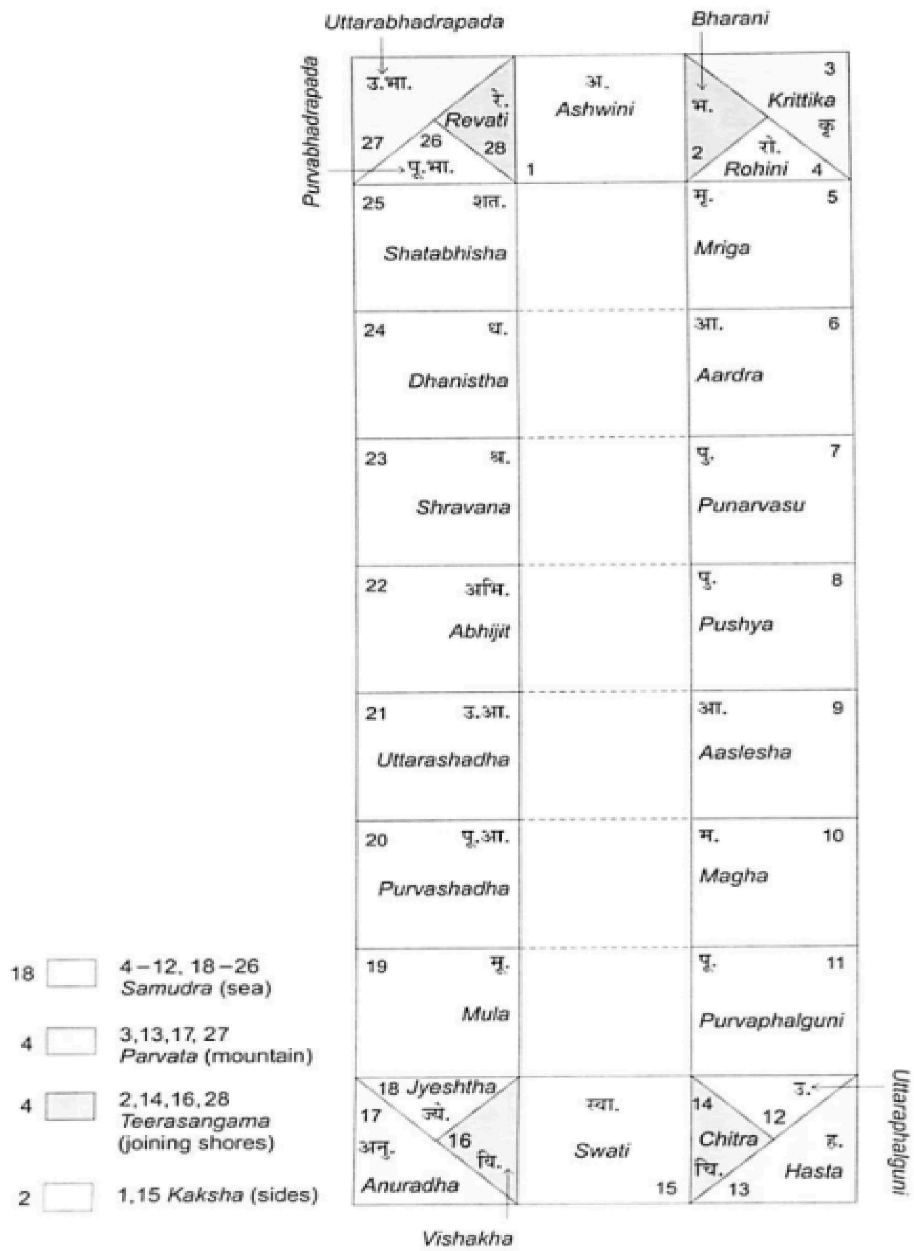
अतिवृष्टिः समुद्रे स्यादनावृष्टिस्तु पर्वते।
कक्षयोश्चित्तला वृष्टिः सुवृष्टिस्तीरसङ्गमे ॥५६॥

[11] The Viṣuva may happen twice, either in the Meṣa Saṁkrānti or the tulā Saṁkrānti (vernal or autumnal equinox). The third line has not been translated into the DPS ed.- 'If the Viṣuva happens at the time of sunset, then the crop yield would be half'

[12] Note: Draw four' vertical lines to get three vertical 'columns". Three multiplied by the same number is nine. Draw 'ten' horizontal lines over the four vertical lines already drawn in a perpendicular direction so as to get nine horizontal 'columns'. A triangle at each corner of this vertical rectangular figure provides the four 'mountains' required.

[13] [This being the third nakshatra in the list of twenty-eight, the starting number to be written in the northeastern triangle is 'three'. The forecast of rain depends on the transition of the Sun through the sign of Aries coinciding with the nakshatras so grouped-Meshasankramana.]

If the Meṣa-Saṁkrānti happens in the (Samudra) Rohiṇī to Uttaraphālgunī and Jyeṣṭhā to pūrva-bhādrapada, the rain would be much more; if in the (parvata) kṛttikā, hasta, anurādhā and uttarabhādrapada, then the rain won't happen; if in the Both of Kakṣas (Kakṣayoḥ), the rain would be moderate, and if in the (Tīra/Taṭa), then a sufficient amount of rain would happen.[14]



[14] Note: The explanation of the words Parvata, Kakṣa and Tīra has been taken from the RCP's Edition of KP

<u>Prediction of the rainfall in the month of *jaiṣṭha* (Based on the astronomical observations)</u>

चित्रास्वातीविशाखासु ज्यैष्ठमासि निरभ्रता।
ताष्वेव श्रावणे मासि यदि वर्षति बासवः।
तदा संवत्सरो धन्यो बहुशस्यफलप्रदः ॥५७॥

If in Jyeṣṭha the sky is bereft of clouds during Citra, Svati, and Visakha nakṣatras and if it rains in Sravaṇa (August) during the same nakṣatras, the year brings blessings in the form of a profuse yield of crops.

ज्यैष्ठादौ सितपक्षे च आर्द्रादिदशऋक्षके।
सजला निर्जला यान्ति निर्जलाः सजला इव ॥५८॥

Wetland gets no rain, and dry land becomes wet if it rains at the beginning of Jyeṣṭha in the bright fortnight during the ten nakṣatras starting with Āardra.

<u>Prediction of the rainfall in the month of *āṣāḍha* (Based on the wind, rain and astronomical observations)</u>

आषाढ्यां पौर्णमास्यां सुरपतिककुभो वाति वातः सुवृष्टिः
शस्यध्वंसं प्रकुर्याद्दहनदिशिगतो मन्दवृष्टिर्यमेन।
नैऋत्यां शस्यहानिर्वरुणदिशि जलं वायुना वायुकोपः
कौवेर्या शस्यपूर्णा प्रथयति नियतं मेदिनीं शम्भुना च ॥५९॥

In Aṣāḍha Purṇimā, the air comes from East, Agni (South-East), South, Nair̥ta (South-west), West, Vāyu (North-West), North, and Īśāṇa (North-East). The results would be accordingly: Good Rain, Destruction of Crops, Poor Rain, Reduction in crops, Rain with thunder, Good Crop.

आषाढस्य सिते पक्षे नवम्यां यदि वर्षति।
वर्षत्येव तदा देवस्तमावृष्टौ कुतो जलम् ॥६०॥

If it rains on the ninth day of the bright half of Aṣāḍha, it rains (throughout the year) without fail, but if it does not rain on that day, then there can be no hope of rain (in that year).

शुक्लाषाढ्यां नवम्यामुदयगिरितटी निर्मलत्वं प्रयाति स्वीयं कायं विधत्ते खरतनुकिरणो मण्डलाकारयोगम्।
जीमूतैर्वेष्टितोऽसौ यदि भवति रविर्गम्यमानेऽस्तशैले तावत्पर्यन्तमेव प्रगलति जलदो यावदस्त तुलायाः ॥६१॥

If on the ninth day of the bright half of Aṣāḍha, the eastern (Udayagiritaṭa[15]) quarter becomes clear and the Sun with its scorching rays has its normal round-shaped orb and is surrounded by clouds till the proceeding towards the mountain of setting, clouds pour until the Sun's transition through Tula (Libra) is complete.

<u>Prediction of the rainfall in the month of *śrāvaṇa* (Based on the rain and astronomical observations)</u>

[15] *Udayagiritaṭa is the East, and Astāśaila is the West.

रोहिण्यां श्रावणे मासि यदि वर्षति वासवः।
तदा वृष्टिर्भवेत्तावद् यावन्नोत्तिष्ठते हरिः ॥६२ ॥

If in Sravana it rains during the Sun's transition through Rohini, it will continue to rain up to the eleventh day of the bright half of Kartika (November), the day on which Lord Vishnu is believed to rise from his sleep (of four months).

कर्कटे रोहिणीऋक्षे यदि वृष्टिर्न जायते।
तदा पराशरः प्राह हा हा लोकस्य का गतिः ॥६३ ॥

If it does not rain in this month, while (the Sun) is in the sign of Cancer (June-July) (the Rohini nakṣatra), Parashara says, "Alas! Oh! Alas! What will be the future of this world?" (The calamity of famine will bring untold misery to the world).

श्रावणे मासि रोहिण्यां न भवेद्वर्षणं यदि।
विफलारम्भसंक्लेशास्तदा स्युः कृषिवृत्तयः॥ ६४ ॥

All the trouble of starting the farming activity is a waste if it does not rain in Sravana while the Sun is in Rohini.

Sudden Rainfall

जलस्थो जलहस्तो वा निकटेऽथ जलस्य वा।
प्रष्टा पृच्छति वृष्ट्यर्थं वृष्टिः संजायतेऽचिरात्॥ ६५ ॥

If someone asks a question about the rain (to the expert, when an expert)in or near the water or putting water in hand, the rain will happen very soon.

उत्तिष्ठत्यण्डमादाय यदा चैव पिपीलिका।
भेकः शब्दायतेऽकस्मात् तदा वृष्टिर्भविष्यति॥ ६६

When the ants come out with their eggs and the frogs suddenly start sounding then, the rain will definitely happen.

विडाला नकुलाः सर्पा ये चान्ये वा विलेशयाः।
धावन्ति शलभा मत्ताः सद्योवृष्टिर्भवेत् ध्रुवम्॥ ६७ ॥

Cats, Bengal mongooses, snakes and Any animal living or burrowing in holes as well as grass-hoppers, when they start running (escapes) like mad, soon the rain will start, definitely.

कुर्वन्ति बालका मार्गे धूलिभिः सेतुबन्धनम्।
मयूराश्चैव नृत्यन्ति सद्योवृष्टिर्भवेद् ध्रुवम्॥ ६८ ॥

When the childrens make a bridge with the dust in the road and the peacock starts dancing, soon the rain will start, definitely.

आघातवातदुष्टानां नृणामङ्गे व्यथा यदि ।
वृक्षाग्रारोहणं चाहेः सद्योवर्षणलक्षणम् ॥ ६६ ॥

If the people (feel) pain from the accident, the impure vāta and the snakes start lifting over the top of the tree, they are an indication of rain soon.

पक्षयोः शोषणं रौद्रे खगानामम्बुचारिणाम् ।
झिन्झीरवस्तथाकाशे सद्यो वर्षणलक्षणम् ॥ ७० ॥

The birds that stay in the water are seen to dry their wings, and the sound of crickets (could be heard) is the intimation that it will rain soon.

Description/details/explanation: Various signs of sudden rain.

- If an expert on rainfall prediction is approached with a query about rain, they are taking a dip in water, holding water, or are in the vicinity of water.
- Ants emerging from an anthill carrying their eggs.
- A sudden croaking of frogs.
- Cats, mongooses, snakes and other creatures which live in holes, as well as grasshoppers moving around freely.
- Children playing on the road, building bridges of mud/dust, and peacocks dancing.
- People suffering from injury or Vata-dośa, complaining of body pain and snakes climbing on the tree tops.
- Water birds drying their wings in the hot sun and crickets chirping in the sky.

Prediction of rainfall with the help of Planet

चलत्यक्ङ्गारके वृष्टिध्रुवा वृष्टिः शनैश्चरे ।
वारिपूर्णा महीं कृत्वा पश्चात् संचरते गुरुः ॥७२ ॥

Rainfall can be predicted when Mars transits from one sign to another, or when Saturn moves from one sign to the other, but Earth is inundated before Jupiter moves.

ग्रहाणामुदये चास्ते तथा वक्रातिचारयोः ।
प्रायो वर्षन्ति हि घना नृपाणामुद्यमेषु च ॥७२ ॥

Clouds often pour down at the rising and setting of planets, as also at their crooked (Vakra) or irregular movements (Aticāra), and also when kings' planets are in the udyamakāla.

चित्रामध्यगते जीवे भिन्नभाण्डमिव स्रवेत्।
ततः स्वातिं समासाद्य महामेघान् विमुञ्चति ॥७३॥

It rains as if a liquid is oozing from a broken pot (cloud-burst) when Jupiter reaches the middle of its course through Citra. Later, when it reaches Svāti, it releases huge clouds (cloudburst!).

पुष्येणोपचितान् मेघान् स्वातिरेका व्यपोहति।
श्रवणे जनितं वर्षं रेवत्येका विमुञ्चति॥ ७४॥

Water filled in clouds in Pushya is released only in Svati, and that which is stored in Sravana is set free in Revati alone.

Indication of Famine

ध्रुवे च वैष्णवे हस्ते मूले शक्रे चरन् कुजः।
सद्यः करोत्यनावृष्टिं कृत्तिकासु मघासु च॥ ७५॥

If the Maṅgal (Kujaḥ) goes into the Uttaraphālgunī, Uttarāṣāḍha, Uttarabhādrapada and Rohinī (Dhruva) as well as the śravaṇa, hasta, mūla, jyeṣṭha, kr̥ttikā, Maghā, then the rains would not happen.

कुजपृष्ठगतो भानुः समुद्रमपि शोषयेत्।
स एव विपरीतस्तु पर्वतानपि प्लावयेत्॥ ७६॥

If the Sun (Bhānu) goes behind Mars and dries out the sea, if the condition is the opposite, then the mountain (region) could also be flooded.

सद्यो निकृन्तयेद् वृष्टिंचित्रामध्यगतो भृगुः।
अङ्गारको यदा सिंहे तदाङ्गरमयी मही॥ ७७॥
स एव रविणा युक्तः समुद्रमपि शोषयेत्॥ ७८॥

If the śukra stays under the Citrā star, then rain will start immediately. When the maṅgal stays in the Siṁha (Leo), it turns the earth into a fireplace, and if it is accompanied by Ravi (Sun), then the sea can also be dried.

Notes:

- Dhruva indicates uttaraphālguni, uttarāṣāḍha and uttara bhādrapada.
- Vaiṣṇava refers to śravana.
- śakra is believed to be the master of Jyeṣṭha.
- The Sun, situated behind Mars, evaporates even the ocean, while in the opposite situation, it drenches the mountains too.
- Additional information/particulars: Obstruction to rain results as soon as Venus reaches the middle.